

Prospective method-selection stress test (v2)

Status: input and analysis plan frozen at commit 8b14325; 15 prescore DeepSeek choices were frozen at bbf5fb7 and provider-visible trace grading passed 15/15. Choice JSON SHA-256: a87cf337fb8a74e943dc6237a6a0ee629d53aa3c0eee1e4b0f41cf1e5b757949. The 15 choices select B2 five times, B1k five times, B1 three times and B0p twice. No partition benchmark outcome was computed before this freeze. The original five-partition run subsequently **failed** at partition 02 because its sole -1 rating cannot be stratified by the official `random_simple` splitter. This is a within-TRANSCRIPT stress test, **not** an independent external-dataset validation or evidence of improved drug efficacy.

Five disjoint sets of disease IDs were deterministically derived from checksum-pinned TRANSCRIPT v2.0.0. The committed `configs/method_selection_partition_v2.json` fixes the data manifest, four eligible methods (B0p, B1k, B1, B2), `random_simple` protocol, 20 outer seeds, five-fold model selection and three independent model choices per partition. `configs/method_selection_partition_v2_cases.json` contains only label-blind input summaries and staged-data paths; it was committed before model selection. Disease groups are disjoint, but drugs and the parent dataset are shared, and outer test items within a group are reused across runs. Do not call the resulting 15 choices 15 independent biomedical validation cohorts.

Run order is binding:

1. Verify the source checksum and stage partition data with `python -m scripts.stage_method_selection_partition_v2`. Do not inspect partition outcomes before choices are frozen.
2. Set `DEEPSEEK_API_KEY` in the local process environment or an ignored repository `.env`, then run `python -m scripts.run_method_selection_partition_v2 --recover-attempt-trace artifacts/reports/traces/20260926T152907345858Z_method_selection_v2_99b0cdaf25da4b2d9e950ede1436c91d.jsonl`. This recovery revalidates all six previous provider responses (including the earlier recovered first choice), in order, and does not resample them. The runner refuses to overwrite its choice file and refuses to run after partition outcome directories exist. Each choice stores a provider-trace SHA-256. Immediately run `python -m evals.grade_method_selection_traces` and require 15/15 provider-visible request/response matches before computing outcomes. Preserve the resulting choice JSON, its SHA-256 and its JSONL/provider traces before running models. Never commit credentials or unreviewed provider traces.
3. Prepare the pinned RECeSS upstream clone (commit a7f11077271cedf3a98a82e3dc74b6fc0e93986e) with `python -m scripts.prepare_recess_official_upstream`. This traced, idempotent step verifies the commit and applies both local patches with `git apply --recount` (their hunk counts require recount). Then use `python -m scripts.run_method_selection_partition_v2_benchmarks` for a read-only preflight. **Only after step 2 has passed 15/15 trace grading**, use the same command with `--execute` to run all five partitions, four methods and 20 seeds through the official wrappers and automatically score them. This entrypoint refuses existing outcome directories and will not run without the frozen choices. The underlying per-case wrapper commands remain documented in the script for manual audit.
4. The batch command automatically scores after all official outputs pass. For an independent recheck, run `python -m evals.score_method_selection_partition_v2 --output artifacts/reports/method_selection_partition_v2_regrade.json`. The scorer refuses any failed choice trace

and verifies frozen file hashes, complete and identical seed order for all four methods, finite official NS-AUC values and exactly three choices per case. It reports chosen method vs fixed B2, seed-paired differences, and an oracle upper bound; no post-hoc best-of-three selection or treating repeats as independent cohorts is permitted.

Every stage writes JSONL trace receipts under ignored `artifacts/`. A trace records visible calls and outputs, not a model's private reasoning. The historical split-level selector had only two decisions; the new paired audit finds +0.039825 NS-AUC over fixed B2 on the reused weak holdout and a tie on random splits, but does not establish generalizable LLM method-selection skill. Even a positive v2 result would require a genuinely independent dataset and prospective replication before such a claim.

Pre-outcome format-only amendment

The first live v2 call (partition 01, repeat 1) returned a valid B2 tool choice, but its 718-character reason exceeded the then-local 200-character cap. The command stopped before saving a choice report; no partition benchmark was run or viewed. The complete, redacted and SHA-256-chained [provider response](#) (file SHA-256 4ed955d92ac30261bbe79233abc6bf6e6117beb472dc5950bbbd1f734121e96e) and [validation-failure trace](#) (SHA-256 e0b3b62fd6e5c3a0a0e36b402a043699b93b3a10d6d4213d2c0b1c2b325a0b3f) were preserved before the cap changed to 1000 characters. The schema, blind input, eligible methods, split, scoring metric and outcome plan did not change. Recovery checks byte-equivalent model request construction and uses the original provider-visible response as the first of 15 choices, avoiding selective redrawing.

The next run recovered that response, then validated four more choices. Its sixth provider response (partition 02, repeat 3) had a 1009-character reason and exceeded the amended 1000-character cap. Again no choice report was saved and no benchmark outcome was run or viewed. The failed attempt trace and all six provider responses were preserved with SHA-256 chains. The reason cap was raised to 4096 characters, well above the two observed responses while the 400-token request limit stayed unchanged. The recovery function replays each exact prior request/response against its frozen blind case, checks already-validated choices and provider hashes, and continues only from choice seven. No methods, blind input, labels, split, scoring plan or provider response were changed or redrawn.

Runner feasibility failure and disclosed amendment

After the choices were frozen, the original official run completed partition 01 then stopped at partition 02 B2. A traced reproduction raised `ValueError: The least populated classes in y have only 1 member ... [-1] inside stanscofi.training_testing.random_simple_split` before that partition produced a score. The original [batch failure trace](#) has SHA-256 2dc668f67e44155de29b8646510c7aea757a0745fedcf740b679528127824661; the [official wrapper failure trace](#) has SHA-256 232468aa0f27c13b918620b8f20d9a02e39cb92d03bb7573936f61cf8ac199af. Explicit negative counts across the five frozen groups are 0, 1, 0, 7 and 3. No NS-AUC values from partition 01 were examined to choose the amendment.

The separately labeled **positive-only feasibility amendment** maps every explicit -1 rating to 0 (unknown) in all five partitions while preserving positives and both feature tables. This is a post-failure data-protocol change, so its eventual score must **not** be presented as the untouched preregistered v2 result. It retains the previously frozen 15 method choices, uses separate staged data and output directories, and still requires identical official runner splits, seeds and five-fold tuning within each partition across B0p/B1k/B1/B2. The staged [amendment manifest](#) has SHA-256 5b512abd97df4de2a85ea821a10ac0e54293d06be13c9e67cfa88abd56f7ce35 and its read-only preflight found no amended outcome directories. It was frozen before any amended run. In a fresh clone, first stage the original partition data, then use `python -m scripts.stage_method_selection_partition_v2_positive_only --verify-existing-manifest` to rebuild the amended data and verify the committed manifest without overwriting it. Run `python -m scripts.run_method_selection_partition_v2_benchmarks --positive-only-manifest artifacts/reports/method_selection_partition_v2_positive_only_manifest.json` for read-only preflight. The scorer writes a distinct amended status and manifest hash; `--execute` is deliberately non-overwriting and requires a fresh outcome directory.

Amended result (descriptive negative finding)

The amended official runner completed all 5 partitions × 4 methods × 20 outer seeds with 5-fold inner tuning and exact seed-order matching. The [scored report](#) has SHA-256 76a29338e361fc5c9f6cfb96d9186c059563d1cda8a66cf28246f1aa1fcf27a3; an independent regrade matched every outcome, source hash and score. The frozen 15 choices passed provider-visible trace grading 15/15, but choosing among methods from label-blind metadata **underperformed** simply fixing B2. No selected non-B2 method beat B2 in its partition: five B2 ties and ten losses across the 15 individual choices. The repeats share outcomes and are not independent datasets.

Partition 01 · B2, B1k, B1

- **Selected mean NS-AUC:** 0.37087
- **Fixed B2 NS-AUC:** 0.46433
- **Paired delta vs B2:** -0.09346
- **Seed wins/ties/losses:** 3/0/17

Partition 02 · B2, B2, B0p

- **Selected mean NS-AUC:** 0.50407
- **Fixed B2 NS-AUC:** 0.50611
- **Paired delta vs B2:** -0.00204
- **Seed wins/ties/losses:** 12/0/8

Partition 03 · B1, B1k, B0p

- **Selected mean NS-AUC:** 0.41324
- **Fixed B2 NS-AUC:** 0.52828
- **Paired delta vs B2:** -0.11504
- **Seed wins/ties/losses:** 2/0/18

Partition 04 · B1k, B2, B1

- **Selected mean NS-AUC:** 0.43971
- **Fixed B2 NS-AUC:** 0.56251
- **Paired delta vs B2:** -0.12280
- **Seed wins/ties/losses:** 0/0/20

Partition 05 · B2, B1k, B1k

- **Selected mean NS-AUC:** 0.30476
- **Fixed B2 NS-AUC:** 0.52298
- **Paired delta vs B2:** -0.21822
- **Seed wins/ties/losses:** 0/0/20

Partition Equal-weight partition mean · —

- **Selected mean NS-AUC: 0.40653**
- **Fixed B2 NS-AUC: 0.51684**
- **Paired delta vs B2: -0.11031**
- **Seed wins/ties/losses:** 0 positive / 5 negative partitions

This is evidence **against** this particular feature-only LLM selector in these amended within-dataset partitions, not evidence that LLMs can never assist with model selection. The altered rating protocol, shared parent dataset, only five disease groups and reused seeds preclude a clean confirmatory or clinical claim. A better next design would let a selector use training-only inner-CV evidence under a newly frozen protocol, then evaluate it on genuinely unseen outer partitions or an external dataset; these five outcomes must not become that future test set.